

MME IPMP Syllabus

IPMP #	Aptitude			Portions			
	Quantitative	Logical Reasoning	Verbal Ability	Management	Core	Analytics	Software
IPMP #1	LCM & HCF, Work and wages, Pipes and Cisterns, Time speed Distance, Train boats streams, Percentages, Profit & Loss, Averages, Age	Missing numbers, Odd one out, Data sufficiency, Assertion and Reasoning, Coding-decoding, Data interpretation (charts, graphs)	Comprehension passage, Error correction, Rearrangement, Cloze test, Jumbles sentences and passages, Antonyms Synonym, Idioms and phrases, Vocabulary and grammar	Guesstimates: Household approach, Time-based approach, Top-down framework VS bottom-up Framework Case Study: Profitability framework, Porter's five force, SWOT analysis, Cost optimization Tech: Introduction to SQL and Databases, Data Types, Tables, and Schema, CRUD Operation, Filtering Data	PTHT, Physical Metallurgy, Mechanical Behavior of materials, Transport Phenomena, Heat and Mass transfer, Strength of materials, Thermodynamics	Pr obability: Basic definitions, Bayes Theorem, Probability Density, Distribution Function, Data distributions (normal, skewed), and other data distribution types. Stats: Descriptive statistics (Measures of Central Tendency, Variability), Normal Distribution, Sampling Methods, Outlier detection (statistical methods) Languages: Basics Of Python (including libraries such as pandas, NumPy, matplotlib, and Seaborn) & SQL ML: Intro to ML and its types, Linear Regression, bias-variance trade-off, Regularization Techniques, Polynomial Regression, Logistic Regression, Data Preprocessing (handling missing values (imputation, Deletion), handling outliers, Feature transformations (encoding techniques, normalization/standardization)), Data Visualization (different plots and their usage)	Detailed contents below
IPMP #2	Ratio Proportion and partnership, Mixtures and Allegations, Algebra, Simple Interest, Compound Interest, Mensuration 2D and 3D, Simplification and Approximation	Syllogism, Alphanumeric series , Blood relations , Deductive reasoning, Data interpretation (Charts and graphs)		Guesstimates: Population Based approach, Geography approach, MECE framework, Pareto framework Case Study: Market entry framework, 4P', Pricing framework, PESTEL Analysis, Customer segmentation Tech: Basic Joins, Subqueries and Nested Queries, Views, Transactions, and Rollbacks, Indexing and Performance Optimization	PTHT, IMSM, Fatigue, Creep and Fracture Mechanics, Welding, casting and forming, Mineral Processing & Metallurgical analysis, Alloy development, Powder metallurgy	Stats: Inferential statistics - Hypothesis Testing (t-test, Z test, chi-squared & ANOVA), Correlation and Covariance Languages: Moderate level difficulty with Python and SQL basics Of Excel ML: SVM, Naive Bayes, Decision Trees, Random Forest, Gradient Boosting, XG boost, cross-validation techniques for model selection (hyperparameter tuning), feature selection, Evaluation metrics (Confusion Matrix, ROC – AUC curve, RMSE, MSE, R2 score)	
IPMP #3	Trigonometry - Height and Distances, Progressions, Logarithms, Permutations and Combination, Probability, Geometry, Race, Clocks and Calendars, Inequalities	Directions, Ranking, Seating arrangements, Grouping and selections, Data interpretation (Charts and graphs)		Guesstimates: Data-centric and income-based approach, Process mapping, Funnel and segmentation framework Case Study: Mergers and acquisitions, Growth strategy frameworks, BCG Matrix, 7S Framework (McKinsey) Tech: Normalization and Keys, Joins, Subqueries, Stored Procedures and Functions	NDT, Corrosion, Automotive materials and basics of mechanics and machine design, Material Science, NFE	ML: Unsupervised Learning - clustering algorithms (K-means, hierarchical, DB Scan, etc.), Elbow method, silhouette score, PCA, Dimensionality Reduction and its importance Languages: Advanced SQL (window functions), Python (OOPS concepts), Advanced Excel DL: (Recommended not compulsory) ANN (backpropagation, activation functions, regularization techniques, etc.), NLP (tokenization, stemming, lemmatization, vectorization techniques)	

